PREREQUISITES

Probability toolkit: **Total prob:** $P(A) = \sum_k P(A|B_k)P(B_k)$. **Bayes:** $P(B_k|A) = P(A|B_k)P(B_k)/\sum_j P(A|B_j)P(B_j)$. **Joint:** $p(x,z) = p(x|z)p(z)$. **Marginal:** $p(x) = \int p(x,z)\,dz$ (or $\sum_z$). **Conditional indep.** given θ: $p(\tilde{x}|\theta,x) = p(\tilde{x}|\theta)$. **Indicator:** $\mathbb{E}[I(A)] = P(A)$.
Tower: $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X|Z)]$. **Total variance:** $\text{Var}X = \mathbb{E}[\text{Var}(X|Z)] + \text{Var}[\mathbb{E}(X|Z)]$. **Linear:** $\text{Var}(aX+bY) = a^2\text{Var}X + b^2\text{Var}Y + 2ab\text{Cov}(X,Y)$; $\text{Var}(t^\top\hat{\theta}) = t^\top Vt$; $\text{Var}(AY) = A\text{Var}(Y)A^\top$.
Square trick (CAVI): $\mathbb{E}[(c-\mu)^2] = (c-\mathbb{E}\mu)^2 + \text{Var}\mu$, so $\mathbb{E}[\mu^2] = (\mathbb{E}\mu)^2 + \text{Var}\mu$.
Jensen: f convex $\Rightarrow \mathbb{E}f(X) \geq f(\mathbb{E}X)$. Log is concave: $\log\mathbb{E}Y \geq \mathbb{E}\log Y$, equal iff Y is constant.
Invariance: MLE of $g(\theta)$ is $g(\hat{\theta})$. **Delta:** $g(\hat{\theta}) \approx \text{N}(g(\theta), g'(\theta)^2 I(\theta)^{-1})$. **Taylor:** $g(Y) \approx g(\mu) + (Y-\mu)g'(\mu)$.
Integrals: $\int_0^\infty x^{a-1}e^{-bx}dx = \Gamma(a)/b^a$; $\int_0^1 x^{a-1}(1-x)^{b-1}dx = B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$; $\Gamma(a+1) = a\Gamma(a)$; $\Gamma(n) = (n-1)!$; $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. **Normal integral:** $\int e^{-\frac{A}{2}\mu^2+B\mu}d\mu = \sqrt{2\pi/A}\,e^{B^2/2A}$.
Inverse: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. **Odds:** $o = \frac{p}{1-p}$, $p = \frac{o}{1+o}$.
Complete the square: $-\frac{A}{2}\mu^2 + B\mu$ is the kernel of $\text{N}(B/A,\ 1/A)$. Course log = ln: press **ln** on the FX-82, not log. Round final answers to 2dp. FX-82: type $(-0.5)^2$ with brackets; $\sum\log a_i$ is one **ln** of the product.

Distributions NOT on formula sheet:

Name	pmf / pdf	Mean; Variance
$\text{Bin}(m,p)$	$\binom{m}{x}p^x(1-p)^{m-x}$	mp; $mp(1-p)$
Geo $x{\geq}1$	$p(1-p)^{x-1}$	$1/p$; $(1-p)/p^2$
Geo $x{\geq}0$	$p(1-p)^x$	$(1-p)/p$; $(1-p)/p^2$
$\text{NB}(r,p)$	$\binom{x+r-1}{x}p^r(1-p)^x$	$r(1-p)/p$; $r(1-p)/p^2$
$\text{Beta}(a,b)$	$\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}$	$\frac{a}{a+b}$; $\frac{ab}{(a+b)^2(a+b+1)}$

Variances the sheet omits: Exp(rate λ) $1/\lambda^2$; Gamma(a, rate b) a/b^2; InvGamma(a,b) $\frac{b^2}{(a-1)^2(a-2)}$ $(a>2)$; Poisson λ.
Gamma extras: $\mathbb{E}[1/X] = b/(a-1)$ $(a>1)$; $\mathbb{E}[\log X] = \psi(a) - \log b$, $\psi = \Gamma'/\Gamma$. Scale form: $x^{a-1}e^{-x/b}/(\Gamma(a)b^a)$ has mean ab. $\tau \sim \text{Gamma}(a,b) \Leftrightarrow 1/\tau \sim \text{InvGamma}(a,b)$; if $\sigma^2 \sim \text{InvGamma}(a,b)$, $\mathbb{E}[1/\sigma^2] = a/b$.
Bivariate Normal: $X_1|X_2{=}x_2 \sim \text{N}(\mu_1 + (x_2-\mu_2)\sigma_{12}/\sigma_2^2,\ \sigma_1^2 - \sigma_{12}^2/\sigma_2^2)$.
Facts: $Z^2 \sim \chi_1^2$; sum of k indep. $Z^2 \sim \chi_k^2$ (mean k, var $2k$); $Z/\sqrt{U/r} \sim t_r$; $(U/m)/(V/n) \sim F_{m,n}$; $t_\nu^2 \sim F_{1,\nu}$. $\sum$ Pois is Pois; $\sum_1^n \text{Exp}(\lambda)$ is $\text{Gamma}(n,\lambda)$.
Critical values: z: 1.645 (90%), **1.96** (95%), 2.576 (99%). $\chi^2_{df}(.95)$: 1: **3.84**, 2: 5.99, 3: 7.81, 4: 9.49, 5: 11.07, 6: 12.59, 10: 18.31, 30: 43.77. $\chi_1^2(.99)$: 6.63.

Confidence intervals: estimate ± critical value × SE. **Say:** "if we repeated this, 95% of such intervals would contain the true value"; not "95% chance θ is here" (that is a credible interval). CI excludes 0 (1 for a ratio) $\Leftrightarrow$ reject H_0 at 5%.

Tests: H_0 no effect. Reject if statistic beyond the cutoff, or $p < \alpha$. **p-value:** P(this extreme or more | H_0), not P(H_0 true). **Type I:** reject true H_0 (α). **Type II:** keep false H_0 (β). **Power:** $1-\beta$. Write "reject" or "insufficient evidence", never "accept". χ^2, F right-tailed.

Estimators: **Bias** $= \mathbb{E}\hat{\theta} - \theta$. **MSE** $= \text{Var}\hat{\theta} + \text{Bias}^2$. **Consistent:** $\hat{\theta} \to \theta$. **SE:** estimated sd of $\hat{\theta}$. **Precision:** 1/variance. $\bar{X} \sim \text{N}(\mu, \sigma^2/n)$ (CLT).

Model terms: **Nested:** A is B with parameters constrained (usually to 0). **Factor**, L levels: $L-1$ dummies vs a reference level. `x1*x2` = `x1+x2+x1:x2`. `I(x^2)`: quadratic. Coefficients mean "holding other predictors fixed". A predictor can flip sign when correlated with others.

LIKELIHOOD & BINOMIAL REGRESSION

Likelihood: independent Y_i with densities $f_i(\cdot;\theta)$ (may depend on covariates): $l(\theta) = \sum_i \log f_i(y_i;\theta)$. MLE $\hat{\theta}$ maximises l.
Score: $U(\theta) = \partial l/\partial\theta$. **Observed information:** $J(\theta) = -\partial^2 l/\partial\theta\,\partial\theta^\top = -H$, depends on y. **Fisher information:** $I(\theta) = \mathbb{E}J(\theta;Y)$, no y. This is the whole sample (n× one-obs info if iid). Bigger information = sharper curvature = smaller variance.
Asymptotics ($n\to\infty$, θ^* true): **Consistent:** $P(|\hat{\theta}-\theta^*| > \epsilon) \to 0$. **Normal:** $\hat{\theta} \approx \text{N}(\theta^*, I(\theta^*)^{-1})$. **Efficient:** asymptotically unbiased with smallest variance $I(\theta^*)^{-1}$.
Regularity: l smooth (3rd derivatives exist, continuous, bounded expectation); support of Y free of θ; parameter space finite-dimensional, free of Y; θ^* not on boundary.
Q1 recipe: (a) l. (b) l', l''; $I = -\mathbb{E}[l'']$, replace $\sum x_i$ by $n\mathbb{E}X$ if data remain. (c) Solve $l' = 0$, check $l'' < 0$, state $\hat{\theta} \approx \text{N}(\theta, 1/I(\theta))$.
Marks: no data in l'': write "$J = I$, no expectation needed".
New parameter: the MLE plugs in (invariance), but I must be redone: one Bin: $I(p) = \frac{m}{p(1-p)}$, $I(\eta) = mp(1-p)$.
Course I is the whole sample: never divide by n again.

iid $X_i \sim$	MLE	$I(\theta)$
Exp(rate λ)	$1/\bar{x}$	n/λ^2
Exp(mean λ)	$\bar{x}$	n/λ^2
Poisson(λ)	$\bar{x}$	n/λ
$\text{N}(\mu_0, 1/\tau)$	$n/\sum(x_i-\mu_0)^2$	$n/(2\tau^2)$
$\text{N}(\mu_0, \sigma^2)$	$\sum(x_i-\mu_0)^2/n$	$n/(2\sigma^4)$
$\text{N}(\mu, \sigma_0^2)$	$\bar{x}$	n/σ_0^2
$\text{Bin}(m,p)$	$\bar{x}/m$	$nm/(p(1-p))$
Geom $x \geq 1$	$1/\bar{x}$	$n/(p^2(1-p))$
Geom $x \geq 0$	$1/(1+\bar{x})$	$n/(p^2(1-p))$
$\text{Pois}(t_i\lambda)$	$\sum x_i/\sum t_i$	$\sum t_i/\lambda$
Gamma(a kn., λ)	$a/\bar{x}$	na/λ^2
$\kappa(1+x)^{-\kappa-1}$	$n/\sum\log(1+x_i)$	n/κ^2
NegBin$(r, 1{-}p)$, one y	$y/(y+r)$	$r/(p(1-p)^2)$

Wald CI for $t^\top\theta$: $\hat{\theta} \approx \text{N}(\theta^*, I(\hat{\theta})^{-1})$, so $t^\top\hat{\theta} \pm z_\alpha\sqrt{t^\top I(\hat{\theta})^{-1}t}$, $\Phi(z_\alpha) = 1-\alpha/2$. Unit vector t: $\hat{\theta}_i \pm z_\alpha\sqrt{[I(\hat{\theta})^{-1}]_{ii}}$. No I? Use $J(\hat{\theta};y)$.
LR confidence region: $\{\theta : 2l(\hat{\theta}) - 2l(\theta) \leq \chi_k^2(1-\alpha)\}$, $k = \dim\theta$. Better than the Normal approximation for small n. θ_0 is inside iff $2[l(\hat{\theta}) - l(\theta_0)] \leq \chi_k^2(.95)$. The Wald box ignores correlation: a point can sit inside both intervals yet outside the region.
Wald test: $H_0: \beta_j = 0$. $z^* = \hat{\beta}_j/\text{se}(\hat{\beta}_j) \sim \text{N}(0,1)$ under H_0. Reject if $|z^*| > 1.96$.
Null not 0: $z = (\hat{\beta} - b_0)/\text{se}$; R's z column only tests 0.
LRT: Reduced R nested in Full F, k fewer parameters. $LR^* = -2[\log L(\hat{\theta}^R) - \log L(\hat{\theta}^F)] \sim \chi_k^2$ under R. If $LR^* > \chi_k^2(1-\alpha)$ reject R, pick F.
Wald vs LRT: asymptotically equivalent ($z^{*2} \approx \chi_1^2$), but the χ^2 approximation to the LR is generally better, so **prefer LRT**.

Binomial regression: $Y_i \sim \text{bin}(m_i, p_i)$ independent, m_i known (binary: $m_i = 1$), $g(p_i) = \eta_i = x_i^\top\beta$. Linear $p = \beta_0 + \beta_1 t$ fails because $0 \leq p \leq 1$ is needed.
Logit: $\eta = \log\frac{p}{1-p}$, $p = \frac{e^\eta}{1+e^\eta} = \frac{1}{1+e^{-\eta}}$. **Probit:** $\eta = \Phi^{-1}(p)$, $p = \Phi(\eta)$. **Cloglog:** $\eta = \log(-\log(1-p))$, $p = 1-e^{-e^\eta}$.
Logistic curve $\eta = \beta_0 + \beta_1 t$: $p = \frac{1}{2}$ at $t = -\beta_0/\beta_1$ (location), slope there $\beta_1/4$ (steepness).

$$l(\beta) = c + \sum_i \left[y_i\log\frac{p_i}{1-p_i} + m_i\log(1-p_i)\right]$$

Logit: $l = c + \sum_i[y_i\eta_i - m_i\log(1+e^{\eta_i})]$. No closed form: use `optim` or IWLS.
Logit information: $J = I = X^\top WX$, $W = \text{diag}(m_ip_i(1-p_i))$ (no y_i left). One predictor: entries $\sum m_ip_iq_i$, $\sum m_ix_ip_iq_i$, $\sum m_ix_i^2p_iq_i$, $q_i = 1-p_i$.
Predict at x_0: $\hat{\eta}_0 = x_0^\top\hat{\beta}$, $\hat{p}_0 = g^{-1}(\hat{\eta}_0)$.
CI for p: p is not linear in β, so (1) CI for η: $\hat{\eta}_0 \pm 1.96\sqrt{x_0^\top Vx_0}$, $V = I(\hat{\beta})^{-1}$; one predictor $x_0^\top Vx_0 = V_{11} + 2x_0V_{12} + x_0^2V_{22}$. (2) Apply g^{-1} to both ends (g monotone).
Two predictors, $x_0 = (1,a,b)$: $x_0^\top Vx_0 = V_{00} + a^2V_{aa} + b^2V_{bb} + 2aV_{0a} + 2bV_{0b} + 2abV_{ab}$. Keep every digit of `vcov`: terms nearly cancel; drop the covariances and se is far too big. Smallest near the centre of the xs, wide when extrapolating.
Odds: $\log o = \eta$. Change x_j by Δ: $OR = e^{\Delta\beta_j}$, CI $\exp\{\Delta(\hat{\beta}_j \pm 1.96\,\text{se})\}$ (swap ends if $\Delta < 0$). Several x change by vector d: $\log OR = d^\top\hat{\beta}$, $\text{se} = \sqrt{d^\top Vd}$.
OR between two non-baseline levels: $\log OR = \hat{\beta}_a - \hat{\beta}_b$, $\text{se} = \sqrt{V_{aa} + V_{bb} - 2V_{ab}}$.
Classify: $\hat{p} \geq$ cutoff; trade off false positives and negatives by cost (screening: fewer false negatives).
Score (logit): $\sum_i(y_i - m_ip_i)x_{ij} = 0$: observed minus fitted.
Links: same predictors with different links are not nested: no LRT, use AIC. They agree mid-range, differ in the tails. Cloglog leaves 0 slowly, nears 1 fast; natural if $p = 1-e^{-\lambda}$ (at least one Poisson event). Probit, cloglog: e^β is not an OR; y stays in the Hessian, so $J \neq I$.
Case-control: logit slopes are still valid log OR; only the intercept shifts, by $\log(\pi_1/\pi_0)$, so no population risks. Not true for probit.
Sparse data (Hauck-Donner): se too big, z too small, a real effect missed: trust ΔD.

GLMs: DEVIANCE, IWLS, DIAGNOSTICS

Deviance: **Saturated S:** one parameter per observation, $\hat{\mu}_i = y_i$. **Null:** intercept only. **Scaled deviance:** $D^A/\phi = -2\log[L(\hat{\beta}_A)/L(S)] = 2(l_S - l_A)$. **Deviance** is $D^A = \sum d_i$. Binomial and Poisson: $\phi = 1$, same thing. $D_{null} - D_{res}$ = improvement from predictors.
$y = 0$: Poisson $d_i = 2\hat{\mu}_i$; binomial $d_i = -2m_i\log(1-\hat{p}_i)$.

Model	d_i $(0\log 0 = 0)$
Normal	$(y_i - \hat{\mu}_i)^2$ (= SSE in LM)
Poisson	$2[y_i\log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)]$
Binomial	$2[y_i\log\frac{y_i}{\hat{\mu}_i} + (m_i - y_i)\log\frac{m_i - y_i}{m_i - \hat{\mu}_i}]$
Gamma	$2[-\log(y_i/\hat{\mu}_i) + (y_i - \hat{\mu}_i)/\hat{\mu}_i]$

Adequacy (goodness of fit), $\phi = 1$ only: H_0: fitted model adequate. If m_ip_i, $m_i(1-p_i) \geq 5$ (binomial) or $\lambda_i \geq 5$ (Poisson) then $D \approx \chi^2_{n-p}$, p including β_0. Reject if $D > \chi^2_{n-p}(.95)$; rough check $D \approx df$. Saturated model has n parameters (not fixed), so ML theory does not apply: D need not be χ^2. Small m_i or counts (binary $m_i = 1$ especially): no adequacy test, ΔD still fine. Normal, Gamma, IG: ϕ estimated, cannot test adequacy. D far above the cutoff: misfit is real even if a few counts are small. 0/1 data: D depends only on $\hat{p}_i$, so says nothing about fit.
Lack of fit may be overdispersion, wrong mean or missing interaction: check residuals vs $\hat{\eta}$ first.
Nested: A in B, s extra parameters. $(D^A - D^B)/\phi = -2[l_A - l_B]$, l_S cancels. $\phi = 1$: $D^A - D^B \approx \chi_s^2$. ϕ unknown: F test (see Overdispersion). Subtract residual deviances, not null deviances. $D_{null} - D_{res}$ on $p-1$ df tests all predictors at once. **Numeric score vs factor** (L levels), or a group-level covariate vs a group factor: nested, $s = L-2$.
AIC (non-nested too): $AIC = 2p - 2\hat{l}$, smaller is better, same data only, not a test. B has s more parameters: $AIC_B - AIC_A = 2s - (D_A - D_B)/\phi$ (exact for $\phi = 1$). An extra parameter pays only if it cuts D by more than 2.
From R output: $p = n-$residual df; n = null df $+1$. $\phi = 1$: $\hat{l} = p - AIC/2$, $l_S = \hat{l} + D/2$ (same l_S for every model on those data). Gamma, Normal: R's AIC counts ϕ, so $\hat{l} = (p+1) - AIC/2$. Check: two models on the same data give the same l_S. Continuous data: $\hat{l} > 0$, AIC < 0 are fine.

Exponential family: $f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right]$. θ canonical (location), ϕ dispersion (scale). Form not unique. Convention $a(\phi) = \phi$.
$\mathbb{E}Y = b'(\theta)$, $\text{Var}Y = b''(\theta)a(\phi) = v(\mu)a(\phi)$. **Variance function** $v(\mu) = b''((b')^{-1}(\mu))$: how spread changes with mean. **Canonical link** $g = (b')^{-1}$, $g(\mu) = \theta$. Canonical: $g'(\mu) = 1/v(\mu)$.
Proof: $\int f\,dy = 1$. Differentiate in θ: $\mathbb{E}[(Y-b')/a] = 0$, so $\mathbb{E}Y = b'$. Again: $\mathbb{E}[(Y-b')^2/a^2] - b''/a = 0$, so $\text{Var}Y = b''a$.
Recipe: write $\log f$; find y×(something) $= \theta/a$; the term with no y gives $-b/a$; leftovers c. Write b in θ only. Check $b'(\theta) = \mu$. State all of θ, ϕ, a, b, c and write "support free of θ". Not exp family if the support depends on θ, or the parameter multiplies y^2, $\log y$ or e^y rather than y.

- **Normal:** $\theta = \mu$, $\phi = \sigma^2$, $b = \theta^2/2$, $c = -\frac{1}{2}(\frac{y^2}{\phi} + \log 2\pi\phi)$; $g = \mu$, $v = 1$.
- **Poisson:** $\theta = \log\lambda$, $\phi = 1$, $b = e^\theta$, $c = -\log y!$; $g = \log\mu$, $v = \mu$.
- **Binomial:** $\theta = \log\frac{p}{1-p}$, $\phi = 1$, $b = m\log(1+e^\theta)$, $c = \log\binom{m}{y}$; $g = \log\frac{\mu}{m-\mu}$, $v = \mu(1-\mu/m)$.
- **Gamma**(ν,λ): $\theta = -\lambda/\nu = -1/\mu$, $\phi = 1/\nu$, $b = -\log(-\theta)$, $c = (\frac{1}{\phi} - 1)\log y - \frac{\log\phi}{\phi} - \log\Gamma(\frac{1}{\phi})$; $g = -1/\mu$ (R: $1/\mu$), $v = \mu^2$, $\text{Var}Y = \phi\mu^2$.
- **Exponential**(λ): $\theta = -\lambda$, $\phi = 1$, $b = -\log(-\theta)$, $c = 0$; $g = -1/\mu$, $v = \mu^2$. $\mu = -1/\eta$ needs $\eta < 0$, so canonical link can fail: use log. ϕ known: χ^2 tests.
- **Inverse Gaussian:** $\theta = -\frac{1}{2\mu^2}$, $\phi = 1/\lambda$, $b = -\sqrt{-2\theta}$, $c = -\frac{1}{2}[\log(2\pi\phi y^3) + \frac{1}{\phi y}]$; $g = -\frac{1}{2\mu^2}$ (R: $1/\mu^2$), $v = \mu^3$.
- **Weibull,** k known: $Y = X^k \sim$ Exp(rate λ^k).
- **Geometric,** $y \geq 0$: $\theta = \log(1-p)$, $\phi = 1$, $b = -\log(1-e^\theta)$, $c = 0$; $v = \mu + \mu^2$.
- **NegBin,** r known: $\theta = \log\frac{\mu}{\mu+r}$, $\phi = 1$, $b = -r\log(1-e^\theta)$, $c = \log\frac{\Gamma(y+r)}{\Gamma(r)y!}$; $v = \mu + \mu^2/r$. r unknown: not exp family.
- **Proportion** $Y = X/m$: θ logit, $b = \log(1+e^\theta)$, $a(\phi) = 1/m$, $v = \mu(1-\mu)$ (R's scale). **Mean of k Poissons:** $b = e^\theta$, $a(\phi) = 1/k$, $\text{Var}Y = \mu/k$; ϕ known from the design.

GLM: Y from exponential family and $\mu = \mathbb{E}Y = g^{-1}(x^\top\beta)$; g monotone differentiable link. We model location $\eta = x^\top\beta$; scale sorts itself out. Canonical link: $\theta_i = x_i^\top\beta$.

Fitted mean: log link e^η; logit $me^\eta/(1+e^\eta)$; identity η; inverse $1/\eta$.
Newton-Raphson: $x_{n+1} = x_n - f'(x_n)/f''(x_n)$; $\theta^{n+1} = \theta^n - H(\theta^n)^{-1}U(\theta^n)$. **Fisher scoring:** replace J by I (positive definite): $\theta^{n+1} = \theta^n + I(\theta^n)^{-1}U(\theta^n)$. Equal under a canonical link.
Score: $U_j = \sum_i \frac{(y_i - \mu_i)x_{ij}}{a(\phi)v(\mu_i)g'(\mu_i)}$. Canonical link: $X^\top(y - \hat{\mu}) = 0$, so $\sum\hat{\mu}_i = \sum y_i$ with an intercept.
Canonical link: $W = v(\mu)$, $J = I$, NR = Fisher scoring = IWLS, $X^\top y$ sufficient, and fitted totals match the data: group totals, $\sum x_{ij}\hat{\mu}_i = \sum x_{ij}y_i$. Non-canonical: J contains y; same $\hat{\beta}$.
WLS: $Y = X\beta + \varepsilon$, $\varepsilon \sim \text{N}(0,\Sigma)$: $\hat{\beta} = (X^\top\Sigma^{-1}X)^{-1}X^\top\Sigma^{-1}y$. **Idea:** $g(Y_i) \approx Z_i = g(\mu_i) + (Y_i - \mu_i)g'(\mu_i)$ with $\text{Var} = g'(\mu_i)^2v(\mu_i)a(\phi)$; depends on β, so iterate.
IWLS: (1) $\hat{\mu}^0 = y$. (2) $z_i = g(\hat{\mu}_i) + (y_i - \hat{\mu}_i)g'(\hat{\mu}_i)$, $W_{ii} = 1/[g'(\hat{\mu}_i)^2v(\hat{\mu}_i)]$. (3) $\hat{\beta} = (X^\top WX)^{-1}X^\top Wz$, $\hat{\mu}_i = g^{-1}(x_i^\top\hat{\beta})$. (4) Repeat until $\hat{\beta}$ stable. $a(\phi)$ cancels. Same as Fisher scoring. Needs only g, v.
One step by hand: $\hat{\mu}^0 = y$, so $z_i = g(y_i)$, $W_{ii} = 1/[g'(y_i)^2v(y_i)]$ (Poisson log: $\log y$, y). $X^\top WX = \begin{pmatrix} \sum W & \sum Wx \\ \sum Wx & \sum Wx^2 \end{pmatrix}$, $X^\top Wz = \begin{pmatrix} \sum Wz \\ \sum Wxz \end{pmatrix}$. $y = 0$ breaks the start: R uses $y + .1$ (Poisson), $(y+.5)/(m+1)$ (binomial).

Family, link	z_i	W_{ii}
Poisson, log	$\eta + (y-\mu)/\mu$	μ
Binom, logit	$\eta + \frac{(y-\mu)m}{\mu(m-\mu)}$	$\mu(m-\mu)/m$
Gamma, log	$\eta + (y-\mu)/\mu$	1

Other W: Poisson $\sqrt{\mu}$: 4. Gamma $1/\mu$: μ^2. Gamma identity: $1/\mu^2$ $(z = y)$. IG $1/\mu$: μ. Cloglog: $m(1-p)L^2/p$, $L = e^\eta$.
Variance: $\text{Var}\hat{\beta} = (X^\top\hat{\Sigma}^{-1}X)^{-1}$, $\hat{\Sigma}_{ii} = g'(\hat{\mu}_i)^2v(\hat{\mu}_i)a(\hat{\phi})$. Same as $a(\hat{\phi})(X^\top\hat{W}X)^{-1}$: $a(\phi)$ does not cancel here. Gamma log: $\hat{\phi}(X^\top X)^{-1}$. ϕ known but R took 1: SE×$\sqrt{a(\phi)}$.
Estimate ϕ: $\hat{\phi} = X^2/(n-p)$, Pearson $X^2 = \sum_i(y_i - \hat{\mu}_i)^2/v(\hat{\mu}_i)$. Consistent. Not $D/(n-p)$. If $\phi = 1$ and the model is adequate, $X^2 \approx \chi^2_{n-p}$.
Rate model: $Y_i \sim \text{Pois}(\lambda_i)$, $\lambda_i \propto t_i$: $\log\lambda_i = \log t_i + x_i^\top\beta$. `offset(log(t))` fixes the coefficient of $\log t$ at 1 (uses no df). e^β = rate ratio; $\hat{\lambda} = te^{x^\top\hat{\beta}}$. **Check it:** free coefficient of $\log t$, $z = (\hat{\beta} - 1)/\text{se}$, or ΔD on 1 df. Forget $\log t$ in $\hat{\eta}$ and you get the rate, not the count.
Gamma GLM: `Gamma(link="log")` etc. ϕ estimated, so t and F. $\text{sd}/\mu = \sqrt{\phi}$. Pick link by plotting $g(y)$ vs x.
Intercept only: $\hat{\mu} = \bar{y}$, $\hat{\beta}_0 = g(\bar{y})$; Poisson log se $= 1/\sqrt{\sum y_i}$. **Quadratic:** turning point $x = -\hat{\beta}_1/(2\hat{\beta}_2)$. **Log link with** $\log x$: $\mu = e^{\beta_0}x_1^{\beta_1}x_2^{\beta_2}$; test a theory value by $(\hat{\beta} - b_0)/\text{se}$. **Identity link:** CI directly on the response scale. **Inverse link:** larger η means smaller μ.

Residuals: **Response:** $y_i - \hat{\mu}_i$; not homoskedastic unless v constant. **Pearson:** $r_P(i) = (y_i - \hat{\mu}_i)/\sqrt{v(\hat{\mu}_i)}$, $\sum r_P^2 = X^2$. **Deviance:** $r_D(i) = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$, $\sum r_D^2 = D$ (R default). Binomial $v = m\hat{p}\hat{q}$. $y = 0$: $r_D = -\sqrt{2\hat{\mu}}$, a smooth curve on the plot. `type="response"` on `cbind` gives $y/m - \hat{p}$.
Leverage: $H = X(X^\top\hat{\Sigma}^{-1}X)^{-1}X^\top\hat{\Sigma}^{-1}$, H_{ii} measures potential influence (unusual x; also depends on IWLS weight), no ϕ. $\sum H_{ii} = p$: rule of thumb, flag $> 2p/n$. Large leverage need not mean actual influence. By hand: $H_{ii} = w_ix_i^\top\widehat{\text{Var}}(\hat{\beta})x_i$ $(\phi = 1)$, w_i the IWLS weight: $m\hat{p}\hat{q}$ (logit), $\hat{\mu}$ (Poisson log), 1 (Gamma log: H depends on x only). Mean leverage p/n. Tiny $\hat{\mu}$: small H even at extreme x.
Studentised: $\text{Var}(Z - \hat{Z}) = (I - H)\Sigma$, so $r_{SP} = r_P/\sqrt{(1-H_{ii})\hat{\phi}}$, $r_{SD} = r_D/\sqrt{(1-H_{ii})\hat{\phi}}$. Unit variance after adjusting for leverage. ±: above/below fit.
Jack-knife: change in $\hat{\mu}_i$ when y_i removed (outliers): $\text{sign}(y_i - \hat{\mu}_i)\sqrt{(1-H_{ii})r_{SD}^2 + H_{ii}r_{SP}^2}$. $\phi = 1$: sign · $\sqrt{r_D^2 + H_{ii}r_P^2/(1-H_{ii})}$.
Cook: change in the whole fit when i removed: $(\hat{\beta}_{(i)} - \hat{\beta})^\top X^\top\hat{\Sigma}^{-1}X(\hat{\beta}_{(i)} - \hat{\beta})/p \approx r_{SP}^2H_{ii}/[p(1-H_{ii})]$. Same as $r_P^2H_{ii}/[p\hat{\phi}(1-H_{ii})^2]$. Flag near 1 or a clear jump. Gamma: keep $\hat{\phi}$.
Half-normal plot: ordered |values| vs $\Phi^{-1}((n+i)/(2n+1))$. Smooth = fine; jump or kink in tail = problem. Need not be straight.
r_D vs $\hat{\eta}$: no trend, even spread is good. Trend or curve: wrong mean. Fan: wrong variance. **Banding:** discrete counts, not a problem; use a smoother. 0/1 data: two curves, nothing to read.
Linearity: plot $g(y)$ or z vs each x_j: want straight. Fix by transforming x, not y (breaks distribution of Y).
Influential point: refit without it, compare $\hat{\beta}$, D, $\hat{\phi}$. Typically keep unless a recording error.
Classify: **outlier:** big jack-knife, ordinary H, coefficients barely move. **High leverage only:** big H, tiny residual, small Cook. **Influential:** big Cook, coefficients move when dropped (needs both leverage and residual).

OVERDISPERSION & QUASI-LIKELIHOOD

Overdispersion: actual $\text{Var}Y_i$ larger than the model says.
Causes: binomial: dependent trials or p changing trial to trial. Poisson: dependent events or non-constant rate. Random rate: $\text{Var}Y = \mathbb{E}\lambda + \text{Var}\lambda > \mathbb{E}\lambda$. Clusters of size k: $\text{Var}Y = (1 + (k-1)\tau^2)mp(1-p)$.
Detect: $D \gg df$, GOF p tiny, $\hat{\phi} \gg 1$ (no universal cutoff). Rule out wrong mean and outliers first. D/df is close to $\hat{\phi}$, but quote Pearson. A free mean for every group and D still big: must be overdispersion. 0/1 data: cannot be detected.
If ignored: $\hat{\beta}$ unaffected, but SEs too small, z too big, p-values too small, CIs too narrow: spurious significant predictors, too big a model. True se $\approx \sqrt{\hat{\phi}}$se, $z \to z/\sqrt{\hat{\phi}}$, $\Delta D \to \Delta D/\hat{\phi}$. $\hat{\phi} > 1$ only ever makes terms less significant; $\hat{\phi} < 1$: SEs too large.
Fixes: quasi-likelihood; different model (Poisson with Gamma rate gives NegBin, $\text{Var} = \mu + \mu^2/\nu$); Bayes: random effect in η.
Quasi-likelihood: specify only $g(\mu_i) = x_i^\top\beta$, $\mathbb{E}Y_i = \mu_i$, $\text{Var}Y_i = \phi v(\mu_i)$; no distribution. Score $u_i = \frac{Y_i - \mu_i}{\phi v(\mu_i)}$: $\mathbb{E}u = 0$, $\text{Var}u = -\mathbb{E}\frac{\partial u}{\partial\mu} = \frac{1}{\phi v}$, as for a real score.

$$Q_i = \int_{y_i}^{\mu_i}\frac{y_i - t}{\phi v(t)}dt, \quad Q = \sum_i Q_i$$

$\partial Q_i/\partial\mu_i = u_i$. Q need not be a real log-density. $\hat{\beta}$ is consistent and asymptotically Normal as for MLE, but less efficient: use the distribution if known.
Quasi-deviance: saturated $Q = 0$, so scaled $D_Q/\phi = -2\sum Q_i$, $D_Q = -2\sum_i\int_{y_i}^{\mu_i}\frac{y_i - t}{v(t)}dt$.
Quasi-Poisson: $v = \mu$, $Q_i = \frac{1}{\phi}[y_i\log\mu_i - \mu_i]$ + const. **Quasi-binomial:** $v = \mu(m-\mu)/m$, $Q_i = \frac{1}{\phi}[y_i\log\frac{\mu_i}{m_i - \mu_i} + m_i\log(m_i - \mu_i)]$ + const. ϕ cancels in IWLS, so same $\hat{\beta}$, $\hat{\mu}$, D as Poisson/binomial.
Numeric Q_i (definite integral, ≤ 0): $v = \mu$: $\phi Q_i = y\log\frac{\mu}{y} - (\mu - y)$. $v = \mu(m-\mu)/m$: $y\log\frac{\mu}{y} + (m-y)\log\frac{m-\mu}{m-y}$. $v = \mu^2$: $1 - \frac{y}{\mu} + \log\frac{y}{\mu}$. $v = \mu^3$: $-\frac{(y-\mu)^2}{2y\mu^2}$.
Other v: $v = 1$: $\phi Q_i = -\frac{1}{2}(y-\mu)^2$, D_Q = RSS, $\hat{\phi} = \hat{\sigma}^2$. D_Q for $v = \mu$, $\mu(m-\mu)/m$, μ^2, μ^3 is the Poisson, binomial, Gamma, IG deviance. $Q_i \leq 0$: integrand opposes the direction of integration. **Why Pearson, not max Q:** $Q = \frac{1}{\phi} \times (\leq 0)$ rises to 0 as $\phi \to \infty$. **Scaled:** $D_Q/\hat{\phi} = -2\sum Q_i \approx n-p$ by construction, so no fit test.
SE: $\text{se}_Q = \sqrt{\hat{\phi}}\,\text{se}_P$, $\hat{\phi}$ of that same model; reverse: $\text{se}_P = \text{se}_Q/\sqrt{\hat{\phi}}$. Estimate never changes. $t = z/\sqrt{\hat{\phi}}$, $z = t\sqrt{\hat{\phi}}$.
One point: $\text{Var}Y_i = \hat{\phi}m\hat{p}\hat{q}$.
Coefficient: $t^* = \hat{\beta}/\text{se}_Q \approx t_{n-p}$; CI $\hat{\beta} \pm t_{n-p}(.975)\text{se}_Q$ (1.96 if no t given), exp for ratio. Prediction is unchanged, only its CI widens: `predict` on the quasi fit includes $\hat{\phi}$ (`residual.scale` $= \sqrt{\hat{\phi}}$); on the Poisson fit multiply se.fit by $\sqrt{\hat{\phi}}$.
Nested F: $F = \frac{(D_A - D_B)/s}{\hat{\phi}} \approx F_{s,n-p_B}$, $\hat{\phi}$ from the BIG model B. No adequacy test, no χ^2, AIC is NA (no likelihood), `step`/`drop1` do not work. Exact for Normal (LM F test), approximate otherwise. Big model's $\hat{\phi}$ not printed: use $D_B/(n-p_B)$. X^2 can fall while $\hat{\phi}$ rises: df fell, the extra terms explain nothing.
New data type: choose link and v from mean-variance (e.g. proportions: logit, $\mu(1-\mu)$), fit IWLS, $\hat{\phi}$ from Pearson. $\hat{\phi} < 1$ (underdispersion) is allowed. iid mean λ, var $\phi\lambda$: $\hat{\lambda} = \bar{y}$, $\hat{\phi} = \frac{\sum(y_i - \bar{y})^2}{\bar{y}(n-1)}$. Non-integer fractions work in `quasibinomial`. Lognormal: $v = \mu^2$, $\phi = e^{\sigma^2} - 1$.

READING R OUTPUT

`summary(glm)` line by line:

- **Deviance Residuals** (old R): 5-number summary of r_D.
- **Estimate** $\hat{\beta}_j$: change in η per unit of x_j, others fixed. Row `schoolB` = level B vs baseline; `a:b` = interaction; `(Intercept)` = η at $x = 0$, baseline levels. **Std. Error:** $\sqrt{V_{jj}}$.
- **z value** (t if ϕ estimated) = Est/SE: tests $\beta_j = 0$ with every other term kept in. **Pr(>|z|):** two-sided p. Stars `***` $<.001$, `**` $<.01$, `*` $<.05$, `.` $<.1$.
- **Dispersion line:** "taken to be 1" is an assumption, not an estimate. Other value: $\hat{\phi} = X^2/(n-p)$.
- **Null deviance** on $n-1$: intercept-only fit, same for every model on those data. **Residual deviance** on $n-p$: your model.
- **AIC:** $2p - 2\hat{l}$; NA for quasi. **Fisher Scoring iterations:** IWLS steps. **"k observations deleted":** n is smaller than the data.

Recover anything hidden: n = null df $+1$. $p = n-$ res df. Non-intercept terms = null df − res df. SE = Est/z; Est = zSE. $\hat{l} = p - AIC/2$ (Gamma $p+1$); $l_S = \hat{l} + D/2$; $\hat{l}_B = \hat{l}_A + \Delta D/2$. s = res df$_A$ − res df$_B$. $\Delta D = D_A - D_B$. $AIC_B - AIC_A = 2s - \Delta D$. $\hat{\phi} = X^2/(n-p) = (\text{se}_Q/\text{se}_P)^2 = (z/t)^2$; $X^2 = \hat{\phi}(n-p)$. $F = (\Delta D/s)/\hat{\phi}$, so $\hat{\phi} = (\Delta D/s)/F$. $z^2 \approx \Delta D$ and $t^2 \approx F$ on 1 df. Logit $\hat{p} = 1/(1+e^{-\hat{\eta}})$; log link $\hat{\mu} = e^{\hat{\eta}}$. $|z| > 1.96$: $p < .05$; $|z| > 2.58$: $p < .01$.

`anova(A,B,test="Chisq")`, two models: row 1 is the smaller model A, row 2 the bigger B.

- **Resid. Df:** $n - p_A$, $n - p_B$. **Resid. Dev:** D_A, D_B.
- **Df** (row 2): s. **Deviance:** $\Delta D = D_A - D_B \geq 0$.
- **Pr(>Chi):** $P(\chi_s^2 > \Delta D)$. Small: reject A, keep B. Large: keep A.
- `test="F"` adds **F** $= (\Delta D/s)/\hat{\phi}_B$ and **Pr(>F)** on $(s, n-p_B)$.

`anova(m,test="Chisq")`, one model: "terms added sequentially (first to last)".

- **NULL row:** Resid. Df $n-1$, Resid. Dev D_{null}.
- **Term rows:** that term added after those above it. Df its df; Deviance the drop in D; Resid. Df, Resid. Dev running totals.
- Order matters: only the last row tests a term given all the others.
- **Checks:** Df column sums to $p-1$; Deviance column to $D_{null} - D_{res}$; last Resid. Dev is the summary's.
- Read upward: interaction first. If `A:B` is significant keep `A` and `B`, whatever their rows say.

`drop1(m,test="Chisq")`: `<none>` is m; each row drops only that term, so order is irrelevant. Deviance = D after dropping; LRT = rise in D; 1-df term: LRT $\approx z^2$.
`step(m)`: `Start: AIC=` is the current model. Rows sorted by AIC, lowest first: `- x` drops x, `<none>` changes nothing. Top row wins; stops when `<none>` is on top. Last block's formula is the final model. Fails on quasi (AIC NA).

Which test goes with which family:

	Poisson, binomial	Gamma, IG	Quasi
ϕ	1, fixed	$\hat{\phi}$ Pearson	$\hat{\phi}$ Pearson
Coef	z	t_{n-p}	t_{n-p}
Nested	ΔD vs χ_s^2	$F_{s,n-p_B}$	$F_{s,n-p_B}$
Fit	D vs χ^2_{n-p}, fitted ≥ 5	none	none
$\hat{l}$	$p - AIC/2$	$p+1-AIC/2$	AIC NA

Degrees of freedom: **null** $n-1$. **Residual, GOF χ^2, Wald t, $\hat{\phi}$:** $n-p$, p counting the intercept. **All terms:** $p-1$. **LRT χ^2:** s = difference in residual df. **Quasi, Gamma F:** $(s, n-p_B)$. **Numeric term** or `I(x^2)`: 1. **Offset:** 0. **Factor:** $L-1$. **Two factors** `A:B`: $(L_A-1)(L_B-1)$. **Factor by numeric** `A:x`: $L-1$. `A*B`: main effects plus interaction. **Numeric to factor:** $L-2$. **LR region:** $\dim\theta$. **Saturated:** 0 residual df. **Log-linear table:** cells − parameters; `A+B+C` = mutual independence, `A*B` = interaction. **Mixture:** $K-1$ + component parameters.

Interpretation lines: **Log link:** "Holding other predictors fixed, a 1-unit rise in x multiplies the expected count by $e^{\hat{\beta}}$, a $100(e^{\hat{\beta}} - 1)$% change." **Logit:** same for the odds, not the probability. **Identity:** adds $\hat{\beta}$. **Inverse:** $1/\mu$ rises by $\hat{\beta}$, so $\hat{\beta} > 0$ lowers μ. **Factor:** "relative to the reference level". $\log x$ **with log link:** doubling x multiplies μ by $2^{\hat{\beta}}$. **Interaction:** the effect of one predictor depends on the other; a cell vs baseline is exp(its main effects + interaction); add the intercept for the cell mean. **Intercept:** $e^{\hat{\beta}_0}$ = baseline mean, rate or odds at $x = 0$. **Δ units:** $e^{\Delta\hat{\beta}}$, not $\Delta e^{\hat{\beta}}$. CI: exponentiate the ends, never the se. Observational: "associated with", not "causes".

Hard subquestions (R output):

- **Which model, 2 quantitative reasons:** smaller AIC, and $\Delta D > \chi_s^2(.95)$ rejects the smaller one.
- **2 reasons overdispersed:** $D \gg df$ and $\hat{\phi} \gg 1$.
- **Why F, t not χ^2, z:** ϕ unknown, estimated by $\hat{\phi}$.
- **Poisson vs quasi-Poisson:** Poisson fixes $\text{Var}Y = \mu$. Quasi keeps the same mean model, sets $\text{Var}Y = \phi\mu$ with ϕ estimated from Pearson and assumes no distribution. Same $\hat{\beta}$; SEs $\times\sqrt{\hat{\phi}}$; tests become t, F; no AIC.
- **Linear assumption evidence:** $\log(y/t)$ vs x straight; interaction plot parallel lines = no interaction.
- **Outlier:** clear point in half-normal Cook/jack-knife plot. Refit, re-check $\hat{\phi}$: overdispersion typically remains.
- **GOF valid?** Check first: `range(fitted(m))` all ≥ 5; never for 0/1.

- **χ^2 says keep, F says drop:** believe F, $\phi = 1$ is false: prefer the smaller model.
- **Which $\hat{\phi}$?** SE: the model holding that coefficient. F: the bigger model.
- **Factor with no significant dummy:** can still matter: one LRT on $L-1$ df.
- **$z^2 \neq \Delta D$, $t^2 \neq F$:** curvature at $\hat{\beta}$ vs the actual drop; close, not equal. Trust ΔD.

Maid Mast Codes PT1: GLM

`glm(cbind(y,m-y)~x, family=binomial, d)`: grouped binomial. `family=poisson`, `quasipoisson`, `quasibinomial`, `Gamma(link="log")`. `subset=-25` drops row 25; `update(m, subset=-7)` refits without row 7.

- `anova(A,B,dispersion=phi,test="F")` on binomial fits: same F as quasi, but p uses $F_{s,\infty}$. `test="F"` on plain Poisson/binomial: $F = \Delta D/s$, ignores overdispersion (R warns).
- `sum(residuals(m,type="pearson")^2)/m$df.residual` $= \hat{\phi}$.
- `pchisq(D,df,lower.tail=FALSE)` $= P(\chi^2_{df} > D)$. `qchisq(.95,df)`, `qf(.95,s,n-p)`, `qt(.975,n-p)`: cutoffs. `predict(m,newdata=list(x=1),type="response")`: $\hat{\mu}$ (`cbind` binomial: $\hat{p}$, multiply by m; same for `fitted`; offset: `newdata` needs t). `type="link",se.fit=TRUE`: $\hat{\eta}$, se. `rstudent` jack-knife (exact if $\phi = 1$); `influence(m)$hat`; `cooks.distance`; `halfnorm`.
- `sum(m*fitted(f)<5)`: rows failing the rule of thumb. `logLik(m)`: $\hat{l}$ with df p (Gamma $p+1$, quasi NA). `cov2cor(vcov(m))`: correlations. `binomial()$variance(mu)`: v on the proportion scale.
- `optim(start,logL,control=list(fnscale=-1))` maximises l; start values matter.
- `summary(m,dispersion=phi)` rescales SEs ($z \div \sqrt{\phi}$, uses Normal). Old papers: `drop1(m,scale=phi,test="F")`, its F uses $D/(n-p)$, not Pearson.
- `vcov(m)` $= V$ (quasi includes $\hat{\phi}$); `rstandard(m)` $= r_{SD}$; `rstandard(m,type="pearson")` $= r_{SP}$; `hatvalues(m)` $= H_{ii}$.
- `z<-predict(m)+(y-mu)/mu`: working response, log link. `w<-mu`: weight. `dnorm`, `rnorm` take sd; `rgamma(n,shape, rate)`.

MIXTURES & EM

Mixture model: (1) $Z_i \sim \text{Categorical}(\pi_1,\dots,\pi_K)$, Z_i latent label, π_k mixture proportions. (2) $X_i|Z_i = k \sim P(X_i|\Theta_k)$ component.
$P(x_i) = \sum_k \pi_k f(x_i;\Theta_k)$, $L(\theta) = \prod_i \sum_k \pi_k f(x_i;\Theta_k)$.
Free parameters: $K-1$ proportions + component parameters. A mixture is not a linear combination: $\pi_a X_a + \pi_b X_b$ of Normals is Normal; a mixture may be bimodal, or unimodal and skewed.
Mean: $\bar{\mu} = \sum \pi_k \mu_k$. **Var:** $\sum \pi_k \sigma_k^2 + \sum \pi_k(\mu_k - \bar{\mu})^2$. Poisson mixtures are overdispersed.
Clustering: $P(Z_i = k|X_i) = \frac{\pi_k f(x_i;\Theta_k)}{\sum_j \pi_j f(x_i;\Theta_j)}$; assign to the largest. Common factors cancel ($x!$, $\sqrt{2\pi}$) but not π_k or unequal $1/\sigma_k$. Poisson, $K = 2$: $P_{i1} = 1/[1 + \frac{\pi_2}{\pi_1}(\frac{\lambda_2}{\lambda_1})^{x_i} e^{-(\lambda_2-\lambda_1)}]$. Normal, equal σ^2, $\mu_1 < \mu_2$: cluster 1 if $x < \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1}$.

EM algorithm: $l(\theta) = \sum_i \log \sum_k \pi_k f$: the sum inside the log blocks a closed form. If Z were known it would be easy.
Incomplete log-lik: $\log P(X|\theta) = \log \sum_Z P(X,Z|\theta)$.
Complete: $\log P(X,Z|\theta)$. **Q:** $Q(\theta,\theta^0) = \mathbb{E}_{Z|X,\theta^0}[\log P(X,Z|\theta)]$.
Steps: (1) Set θ^0; evaluate incomplete log-lik. (2) **E:** compute $P(Z|X,\theta^0)$. (3) **M:** $\hat{\theta} = \arg\max_\theta Q(\theta,\theta^0)$. (4) Evaluate incomplete log-lik; stop if change $< \epsilon$, else $\theta^0 \leftarrow \hat{\theta}$, go to 2.
Deriving Q: write $P_{ik} = P(Z_i = k|X_i,\theta^0)$.

$$P(X,Z|\theta) = \prod_i \prod_k [\pi_k f(x_i;\Theta_k)]^{I(Z_i=k)}$$

$$\log P(X,Z|\theta) = \sum_i \sum_k I(Z_i = k)[\log \pi_k + \log f(x_i;\Theta_k)]$$

$\mathbb{E}[I(Z_i = k)] = P_{ik}$ and only I is random, so

$$Q = \sum_i \sum_k P_{ik}[\log \pi_k + \log f(x_i;\Theta_k)]$$

$K = 2$: $\pi_1 = \pi$, $\pi_2 = 1-\pi$. Normal: $\log f = -\frac{1}{2}\log(2\pi\sigma_k^2) - \frac{(x_i-\mu_k)^2}{2\sigma_k^2}$ (that π is 3.14). Poisson: $x_i \log \lambda_k - \lambda_k - \log x_i!$. Normal answer in full:

$$Q = \sum_i \sum_{k=1}^2 P_{ik}[\log \pi_k - \tfrac{1}{2}\log 2\pi\sigma_k^2 - \tfrac{(x_i-\mu_k)^2}{2\sigma_k^2}]$$

M-step, $N_k = \sum_i P_{ik}$ (effective count):

- **π:** $\frac{\partial Q}{\partial \pi} = \frac{N_1}{\pi} - \frac{N_2}{1-\pi} = 0$, so $\hat{\pi}_k = N_k/n$. $K \geq 3$: Lagrange $Q + \nu(1-\sum \pi_k)$ gives $\pi_k = N_k/\nu$; sum over k: $\nu = n$.
- **Normal:** $\hat{\mu}_k = \sum_i P_{ik} x_i/N_k$, $\hat{\sigma}_k^2 = \sum_i P_{ik}(x_i - \hat{\mu}_k)^2/N_k$ (from $\frac{\partial Q}{\partial \sigma^2} = \sum P_{ik}[-\frac{1}{2\sigma^2} + \frac{(x_i-\mu)^2}{2\sigma^4}] = 0$). Mean 0: $\sum P_{ik} x_i^2/N_k$. Shared σ^2: $\sum_i \sum_k P_{ik}(x_i - \hat{\mu}_k)^2/n$.
- **Poisson:** $\hat{\lambda}_k = \sum P_{ik} x_i/N_k$; $Q'' = -\sum P_{ik} x_i/\lambda_k^2 < 0$. **Bin$(m,p_k)$:** $\hat{p}_k = \sum P_{ik} x_i/(mN_k)$. **Exp(rate λ_k):** $\hat{\lambda}_k = N_k/\sum P_{ik} x_i$.

Twists rule: collect every term with the parameter across all components and blocks, differentiate once. Exp(λ), Exp(2λ): $\hat{\lambda} = n/\sum(P_{i1} + 2P_{i2})x_i$. Exposure t_i: $\hat{\lambda}_k = \sum P_{ik} x_i/\sum P_{ik} t_i$. N$(\mu,\sigma^2)$, N$(\mu,9\sigma^2)$: $w_i = P_{i1} + P_{i2}/9$, $\hat{\mu} = \sum w_i x_i/\sum w_i$, $\hat{\sigma}^2 = \sum w_i(x_i - \hat{\mu})^2/n$.
n' points known from component j (not drawn from the mixture): no Z, so they add $\sum \log f_j(x_i)$ to Q with weight 1 and no π. $\hat{\pi}_k = N_k/n$, unlabelled n only. $\hat{\mu}_j = (\sum P_{ik} x_i + \sum_{lab} x_i)/(N_j + n')$; same for σ_j^2 with $(x_i - \hat{\mu}_j)^2$. Incomplete log-lik: $\sum_{unlab} \log \sum_k \pi_k f_k + \sum_{lab} \log f_j$.
Observed labels from the mixture: count in both: $\hat{\pi}_k = (N_k + m_k)/(n+m)$. **Known parameters:** do not differentiate. **Tied:** N$(\mu,1)$,N$(-\mu,1)$: differentiate once, $\hat{\mu} = \sum(P_{i1} - P_{i2})x_i/n$. **Zero-inflated:** $P_{i1} = 0$ if $x_i > 0$; $\hat{\lambda} = \sum x_i/(n - N_1)$. $K = 3$: $\pi_3 = 1 - \pi_1 - \pi_2$, three terms in the denominator.
Behaviour: incomplete log-lik **never decreases** (a drop means a bug). Local maxima: sensitive to initial values, so run many starts, keep the **highest incomplete log-lik**.
Label switching: same log-lik, permuted labels, same fit. Equal starts ($\lambda_1^0 = \lambda_2^0$) get stuck at $\hat{\lambda}_k = \bar{x}$, π unchanged.
By hand: table of x_i, P_{i1}, P_{i2}; check $N_1 + N_2 = n$. M-step is the complete-data MLE with hard counts replaced by weights P_{ik}. To choose between runs, plug each $\hat{\theta}$ into $\sum_i \log(\pi f_1(x_i) + (1-\pi)f_2(x_i))$. Keep $\log x_i!$ when reporting Q or log-lik values.
Proof (Jensen): $q(Z) = P(Z|X,\theta^0)$:

$$l(\theta) = \log \sum_Z q \frac{P(X,Z|\theta)}{q} \geq \sum_Z q \log \frac{P(X,Z|\theta)}{q}$$

$= Q(\theta,\theta^0) + C$, $C = -\sum q \log q$. Equality at θ^0, as $P(X,Z|\theta^0)/q = P(X|\theta^0)$ is constant. M-step: $Q(\hat{\theta},\theta^0) \geq Q(\theta^0,\theta^0)$. So $l(\hat{\theta}) \geq Q(\hat{\theta},\theta^0) + C \geq Q(\theta^0,\theta^0) + C = l(\theta^0)$.

EM shortcuts: write "the terms of Q involving λ are…" before each derivative.
Any two classes, no underflow: $P_{i1} = 1/(1+e^d)$, $d = \log \frac{\pi_2\sigma_1}{\pi_1\sigma_2} + \frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2}$.

- Pois(λ), Pois(3λ): $P_{i1} = 1/[1 + \frac{1-\pi}{\pi} 3^{x_i} e^{-2\lambda}]$; $\hat{\lambda} = \sum x_i/\sum(P_{i1} + 3P_{i2})$, known block adds 3 each below.
- Pois($t\lambda_1$), Pois($t\lambda_2$): $P_{i1} = 1/[1 + \frac{1-\pi}{\pi}(\frac{\lambda_2}{\lambda_1})^{x_i} e^{-t_i(\lambda_2-\lambda_1)}]$.
- N(μ,σ^2), N$(-\mu,\sigma^2)$: $P_{i1} = \pi/[\pi + (1-\pi)e^{-2\mu x_i/\sigma^2}]$; $\hat{\mu} = \frac{1}{n}\sum(2P_{i1} - 1)x_i$; $\hat{\sigma}^2 = \frac{1}{n}\sum x_i^2 - \hat{\mu}^2$.
- Zero-inflated Poisson: $x_i = 0$: $P_{i1} = \pi/[\pi + (1-\pi)e^{-\lambda}]$; $x_i > 0$: 0. Binomial: $\pi/[\pi + (1-\pi)(1-p)^{m_i}]$; $\hat{p} = \sum x_i/\sum P_{i2} m_i$.

Mixture moments: $K = 2$ between term $= \pi_1\pi_2(\mu_1 - \mu_2)^2$. Poisson: Var$X = \bar{\lambda} + \pi_1\pi_2(\lambda_1 - \lambda_2)^2$, so intercept-only $\hat{\phi} \approx$ Var$X/\mathbb{E}X$. Exposure t: $t\bar{\lambda} + t^2\pi_1\pi_2(\lambda_1 - \lambda_2)^2$. Zero-inflated: $\mathbb{E}X = (1-\pi)\lambda$, Var$X = \mathbb{E}X(1+\pi\lambda)$.
Cutoffs: Poisson, $\lambda_1 < \lambda_2$: cluster 1 if $x < \frac{\log(\pi_1/\pi_2)+\lambda_2-\lambda_1}{\log(\lambda_2/\lambda_1)}$. Same mean, $\sigma_1 < \sigma_2$: cluster 2 if $|x| > c$, $c^2 = \log \frac{\pi_1\sigma_2}{\pi_2\sigma_1} / (\frac{1}{2\sigma_1^2} - \frac{1}{2\sigma_2^2})$. Misclassified: $\pi_1 P(x > c|1) + \pi_2 P(x \leq c|2)$.
By hand: $\sum P_{i2} x_i = \sum x_i - \sum P_{i1} x_i$. Pooled $\hat{\sigma}^2 = (N_1\hat{\sigma}_1^2 + N_2\hat{\sigma}_2^2)/n$. Variance update uses the new mean.
Choose K: l always rises with K: use AIC, $p = K - 1$ + component parameters (known values not counted).

Maid Mast Codes PT2: EM

`Z=sample(1:K,n,replace=TRUE,prob=pi); X=rnorm(n, mu[Z],sd[Z])`: simulate.

- `L[,k]=dnorm(X,mu[k],s)*w[k]`: $n \times K$ matrix $P(X_i,Z_i{=}k)$.
- `P_ik=prob.x.z/rowSums(prob.x.z)`: E-step.
- `w.new=colSums(P_ik)/sum(P_ik)`: $\hat{\pi}_k$; `sum(P_ik)` $= n$.
- `mu.new=colSums(P_ik*X)/colSums(P_ik)`: $\hat{\mu}_k$.
- `ill=sum(log(rowSums(prob.x.z)))`: incomplete log-lik.
- `delta.ll`, `epsilon`: gain, tolerance. Negative `delta.ll` also stops the loop, so a bug can look converged. Bugs: `/colSums` in E-step; `sum(log(prob.x.z))`.
- `cluster=apply(P_ik,1,which.max)`: largest P_{ik}. `table(true=Z,cluster)`: diagonal correct, needs simulated labels; permute columns if labels switched.
- `length(log_liks) = max.iter+1`: stopped at the cap, not converged, rerun longer. All means equal: every row of `P_ik` is π^0, stuck at $\bar{x}$ after 2 iterations.
- **More bugs:** `pi.new=colSums(P_ik)`: π sums to n, l jumps by $n \log n$ yet still rises. `mu.new=colSums(P_ik*X)/sum(P_ik)`: l falls, loop stops, looks converged.
- **Say:** l rises every iteration and levels off, so it has converged.

BAYES

Bayes: Frequentist: θ fixed, probability = long-run frequency. Bayesian: θ random, probability = degree of belief.

$$p(\theta|x) = \frac{f(x|\theta)p(\theta)}{f(x)} \propto f(x|\theta)p(\theta)$$

Likelihood × prior. Marginal $f(x) = \int f(x|\theta)p(\theta)d\theta$.
Point estimate: posterior mean (minimises squared error loss). **Credible interval:** central 95% of the posterior, a direct probability statement about θ. Normal posterior: mean ± 1.96 sd; samples: `quantile(th,c(.025,.975))`.
Kernels: **Gamma**(a,b): $\tau^{a-1}e^{-b\tau}$. **InvGamma**(a,b): $x^{-a-1}e^{-b/x}$. **Normal:** $e^{-\frac{A}{2}\mu^2 + B\mu}$ is N$(B/A,1/A)$. **Beta**(a,b): $\theta^{a-1}(1-\theta)^{b-1}$.
Recipe: joint → keep factors with the parameter → add exponents → name the kernel. Summaries: $\sum(x_i - c)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - c)^2$. $\tau^{n/2}$ is constant for μ but not for τ: keep it. Each prior appears once.

Likelihood (n)	Prior	Posterior
Bin(m,θ)	Beta	Beta$(a + \sum y_i,\ b + nm - \sum y_i)$
Geom ($x{\geq}1$)	Beta	Beta$(a+n,\ b + \sum y_i - n)$
Poisson(λ)	Gamma	Gamma$(a + \sum y_i,\ b+n)$
Exp(rate λ)	Gamma	Gamma$(a+n,\ b + \sum y_i)$
N$(\mu_0,1/\tau)$	Gamma	Gamma$(a + \frac{n}{2},\ b + \frac{1}{2}S)$
N(μ_0,σ^2)	InvG	InvGamma$(a + \frac{n}{2},\ b + \frac{1}{2}S)$

$S = \sum(x_i - \mu_0)^2$. $\mathbb{E}[\sigma^2|x] = b_n/(a_n - 1)$, not $1/\mathbb{E}[\tau|x] = b_n/a_n$. Poisson($t_i\lambda$): rate $b + \sum t_i$.
Normal mean, σ^2 known: prior N$(\mu,\sigma^2/n_0)$ (n_0 = implicit sample size):

$$\theta|x \sim \text{N}\left(\frac{n_0\mu + n\bar{x}}{n_0+n},\ \frac{\sigma^2}{n_0+n}\right)$$

Prior N(m,s^2): precisions add: variance $(\frac{1}{s^2} + \frac{n}{\sigma^2})^{-1}$, mean = variance $\times(\frac{m}{s^2} + \frac{n\bar{x}}{\sigma^2})$. $\bar{X} \sim$ N$(m, \frac{\sigma^2}{n} + s^2)$ marginally.
Posterior mean is a precision-weighted average of prior mean and MLE. Variance is below both prior and MLE variance. $n \to \infty$: $\to$ N$(\bar{x},\sigma^2/n)$, prior forgotten. $n_0 \to 0$: flat prior. Weight on the prior mean $w = \kappa_0/(\kappa_0 + n\tau) = n_0/(n_0+n)$; $\kappa_0 \approx n\tau$ gives the prior half the say.
Posterior predictive: $p(\tilde{x}|x) = \int p(\tilde{x}|\theta)p(\theta|x)d\theta$. Normal: N(post. mean, post. var + σ^2). Simulate: $\theta^{(l)} \sim p(\theta|x)$, then $\tilde{x}^{(l)} \sim p(\tilde{x}|\theta^{(l)})$; average. Known mean, $\tau|x \sim$ Gamma(a^*,b^*): t with $\nu = 2a^*$, location μ_0, scale^2 b^*/a^*. Poisson-Gamma: Var$\tilde{y} = \mathbb{E}[\lambda|y] + $ Var$(\lambda|y)$; $P(\tilde{y} = 0) = (\frac{b_n}{b_n+1})^{a_n}$.
Pseudo-counts: Beta(a,b) acts like $a+b$ prior trials; Beta(1,1) = U(0,1). Poisson-Gamma mean: $\frac{a+\sum y_i}{b+n} = w\frac{a}{b} + (1-w)\bar{y}$, $w = \frac{b}{b+n}$. Var$(\lambda|y) = a_n/b_n^2$. Gamma interval: mean ± 1.96 sd is a shortcut; exact uses quantiles.
Marginals: Beta-binomial $P(X{=}x) = \binom{m}{x}B(a+x,b+m-x)/B(a,b)$; Poisson-Gamma gives NegBin. Poisson-Gamma $p(y) = \frac{\Gamma(a+y)}{\Gamma(a)y!}\frac{b^a}{(b+1)^{a+y}}$, Var $= \mu + \mu^2/a$. X_1,X_2 iid only given λ: Cov $=$ Var$\lambda > 0$.
Priors: vague N$(0,10^4)$, Gamma$(.001,.001)$; **informative**: concentrated. **Improper:** does not integrate to 1, $p(\mu) \propto 1$, $p(\tau) \propto 1/\tau$ (Gibbs: $a = b = 0$); fine if the posterior is proper ($p(\lambda) \propto 1/\lambda$, Poisson: Gamma$(\sum y_i,n)$, needs $\sum y_i > 0$). $p(\mu,\sigma^2) \propto 1/\sigma^2$ goes straight into π in MH. Flat μ, τ known: N$(\bar{x},\frac{1}{n\tau})$. $p(\tau) \propto 1/\tau$, mean known: Gamma$(\frac{n}{2},\frac{S}{2})$. Flat μ and $1/\tau$: $\tau|x \sim$ Gamma$(\frac{n-1}{2},\frac{1}{2}\sum(x_i - \bar{x})^2)$, needs $n \geq 2$. $x_i \sim$ N$(0,\sigma^2)$, $p(\sigma^2) \propto 1/\sigma^2$: InvGamma$(\frac{n}{2},\frac{1}{2}\sum x_i^2)$, mean $\sum x_i^2/(n-2)$ if $n \geq 3$.
BUGS: `dnorm(m,p)`: second argument is a precision: `mu~dnorm(0,.001)` is N(0,1000), vague. Vague prior: expect mean $\approx \bar{y}$, sd $\approx 1/\sqrt{n\tau}$.
Spike prior: γ discrete: $p(\gamma|x) \propto p(x|\gamma)p(\gamma)$; sample γ, then $\mu|\gamma,x$.
Bayes estimator: minimise posterior expected loss; squared error gives posterior mean. **Bayesian GLM:** vague priors $\beta_j \sim$ N$(0,10^4)$; overdispersion: add $\varepsilon_i \sim$ N$(0,1/\tau)$ to η_i, or $\lambda_i \sim$ Gamma per unit.

SIMULATION

Simulation: **Monte Carlo:** $\frac{1}{N}\sum g(X_j) \approx \mathbb{E}g(X)$; proportions estimate probabilities. `set.seed` makes it repeatable. Need $N \geq \hat{p}(1-\hat{p})/\text{se}^2$ (at most $.25/\text{se}^2$). sd of draws is the posterior spread, fixed; MC s.e. = sd$/\sqrt{N}$ shrinks.
Discrete: $U \sim$ U(0,1); `x<-0; while(F(x)<U) x<-x+1`. Ends with $F(x-1) < U \leq F(x)$, so $P(X = x) = F(x) - F(x-1) = p(x)$. **Read a pmf from a loop:** `while(U>G(Y)) Y<-Y+1` means $F = G$, then difference it. `sample(x,n,replace=TRUE,prob=p)`. Expected comparisons: $\sum$ (tests needed) × prob, so test likely values first.
Inversion: $Y = F^{-1}(U)$: $P(Y \leq y) = P(U \leq F(y)) = F(y)$. Needs F^{-1}. Exp(λ): $-\log(1-U)/\lambda$; U(a,b): $a + (b-a)U$; logistic: $\log\frac{U}{1-U}$; Weibull: $(-\log U)^{1/k}/\lambda$. Pdf $2x$ on (0,1): $\sqrt{U}$. e^{-z} on (0,1): $-\log(1 - U(1-e^{-1}))$.
Rejection, uniform: $f \leq k$ on $[a,b]$. (1) $X \sim$ U(a,b), $Y \sim$ U$(0,k)$. (2) If $Y < f(X)$ return X, else go to 1.
General: envelope density h, $k \geq \sup_x f(x)/h(x)$. (1) $X \sim h$. (2) $Y \sim$ U$(0,kh(X))$. (3) If $Y < f(X)$ return X, else go to 1. Works with unnormalised f^* (posterior): $k \geq \sup f^*/h$. Prior as h: $k = \max_\theta p(x|\theta)$, keep if $U \leq p(x|\theta)/k$. k too small biases the output (where $f > kh$ is under-sampled); too big only slows. h tails at least as heavy as f.
Derive output pdf: $P(X \leq x|\text{accept}) = P(X \leq x,\text{accept})/P(\text{accept}) = \int_{-\infty}^x h(t)\frac{f(t)}{kh(t)}dt/\frac{1}{k} = F(x)$.
Efficiency: expected candidates per sample. $P(\text{accept}) = 1/k$ (with f^*: $\int f^*/k$). $N \sim$ Geom$(1/k)$, $\mathbb{E}N = k$ (f^*: $k/\int f^*$). Least rejection: smallest valid k, h close to f. Uniform envelope: rectangle height $\sup f$ on $[a,b]$, so $k = (b-a)\sup f$. Then accept if $U \leq f^*(X)/\max f^*$.
Gamma(shape m, rate λ), $m > 1$, $h = \mu e^{-\mu x}$: f/h peaks at $x = \frac{m-1}{\lambda-\mu}$. Best $\mu = \lambda/m$, $k^* = m^m e^{-(m-1)}/\Gamma(m)$. $k^* = \infty$ if $m < 1$ or $\lambda \leq \mu$.

Maid Mast Codes PT3: Simulation

- `x<-runif(1,a,b); y<-runif(1,0,K); if(y<fx(x)) return(x)`: rejection.
- `X<- -log(runif(1))*m/lambda`: Exp(rate λ/m) by inversion.
- `tha<-rbeta(N,2,3); X<-rbinom(N,10,tha); table(X)/N`: marginal pmf by simulating parameter then data.
- `while(runif(1)>p) Y<-Y+1` from $Y = 0$: failures before first success, $P(Y = y) = p(1-p)^y$.
- `xt<-rbinom(N,10,theta)`: each draw uses its own θ, so `mean(xt>=8)` is predictive. Label then x is a mixture draw.

MCMC

MCMC: build a Markov chain whose stationary distribution is the posterior. Long-run draws are dependent but approximately from the posterior.
Metropolis-Hastings: target $\pi \propto p(\theta)p(x|\theta)$ (constant not needed). **Proposal** $q(\theta,\theta')$: density of new θ' given current θ; must be simulable.
(1) Start $\theta^{(0)}$, $t = 0$. (2) Propose $\theta' \sim q(\theta^{(t)},\cdot)$. (3)

$$A = \min\left(1,\ \frac{\pi(\theta')q(\theta',\theta)}{\pi(\theta)q(\theta,\theta')}\right)$$

(4) With probability A set $\theta^{(t+1)} = \theta'$, else $\theta^{(t+1)} = \theta$ ($U \leq A$). (5) $t \leftarrow t+1$, go to 2.
Random walk: $\theta' = \theta +$ symmetric noise (U$[-\epsilon,\epsilon]$, N$(0,\epsilon^2)$): q cancels, $A = \min(1,\pi(\theta')/\pi(\theta))$. Uphill always accepted, downhill sometimes, so time spent $\propto \pi$. **Independence:** $q(\theta,\theta') = q(\theta')$, so $A = \min\{1,w(\theta')/w(\theta)\}$, $w = \pi/q$. **Walk on** $\log\tau$**:** extra factor τ'/τ.
Numeric A recipe: (i) π = likelihood × prior at new and current. (ii) q ratio: top is density of proposing current from new, bottom new from current. Multi-part proposals multiply: $\tau_n \sim g(\cdot|\tau_c)$, $\mu_n \sim h(\cdot|\mu_c,\tau_n)$ gives $q(c,n) = g(\tau_n|\tau_c)h(\mu_n|\mu_c,\tau_n)$, $q(n,c) = g(\tau_c|\tau_n)h(\mu_c|\mu_n,\tau_c)$. (iii) Cancel common constants only. (iv) $\min(1,\cdot)$. Out of support: $\pi = 0$, $A = 0$. Log form (symmetric q): `A<-min(1,exp(lp(new)-lp(cur)))`; else add log q ratio.
Ledger: one line each: likelihood (keep $\tau^{n/2}$), each prior (keep $\sqrt{\tau}$), each proposal, multiply, $\min(1,\cdot)$. A correct unfinished ledger still scores.
Proposal ratios $q(c|n)/q(n|c)$: Gamma(k, rate k/τ_c): $(\tau_c/\tau_n)^{2k-1}e^{-k\tau_c/\tau_n + k\tau_n/\tau_c}$ ($k = 1$: Exp, mean τ_c). Exp(rate τ_c): τ_n/τ_c. $\mu_n \sim$ N$(\mu_c,1/\tau_n)$: $\sqrt{\tau_c/\tau_n}\,e^{-\frac{1}{2}(\tau_c-\tau_n)(\mu_n-\mu_c)^2}$. Walk on $\log\tau$: τ_n/τ_c. Same variance both ways: 1.
Symmetric? Swap current and new: same formula for every pair. N(θ_c,s^2) yes; N(0,1), Exp(rate $1/\theta_c$) no. Leave q out and it runs silently on the wrong target. `dunif` prior out of range: log is $-$Inf, rejected. Normal step on τ can go negative: $\pi = 0$, wasted. Independence: q tails as heavy as π, else sticky.
Tuning: want large moves and high acceptance; they compete, so aim for intermediate A. Tiny steps: high acceptance, slow crawl. Huge steps: low acceptance, sticky. Both: high autocorrelation, low ESS. Rule of thumb (not slides): 20–50% accepted.
Read π off A: symmetric q and $A = \min(1,r(\theta_n)/r(\theta_o))$ give $\pi \propto r$.

Gibbs sampler: needs only full conditionals. $\theta = (\theta_1,\dots,\theta_k)$. Start $\theta^{(0)}$. Each sweep: $\theta_j^{(t+1)} \sim p(\theta_j|x,\text{latest values of others})$, $j = 1,\dots,k$. Discard burn-in m; keep the rest as a dependent sample; marginals: keep that coordinate.
Use output: $\mathbb{E}[g(\theta)|x] \approx$ average of $g(\theta^{(t)})$ after burn-in; $P(\theta > c|x) \approx$ proportion above c; credible interval from sample quantiles. Apply g to each draw, then average: $\mathbb{E}[\mu\theta]$ = mean of products; $\mathbb{E}\sigma \neq 1/\sqrt{\mathbb{E}\tau}$; mean of ratios $\neq$ ratio of means. Difference: $\delta^{(t)} = \mu_1^{(t)} - \mu_2^{(t)}$, then `mean(d<0)`.
Special case of MH: proposal = full conditional, so $A = 1$. **Limit:** strong correlation between coordinates gives tiny moves and slow exploration. Bivariate, correlation ρ: lag-1 autocorrelation of one coordinate is ρ^2.
Not a named distribution? Finite support: sample with weights (R normalises `prob`). Otherwise an MH step for that coordinate.
Check before dropping a prior: does it contain the parameter you condition on? Scheme answer: start values ($\tau^{(0)} > 0$), each draw with the newest values, drop burn-in, average.
Standard conditionals ($x_i \sim$ N$(\mu,1/\tau)$, $S = \sum(x_i - \mu)^2$):

- $\mu \sim$ N$(m,\frac{1}{\kappa})$: $\mu|\tau,x \sim$ N$(\frac{\kappa m + \tau n\bar{x}}{\kappa + n\tau},\frac{1}{\kappa + n\tau})$.
- $\mu|\tau \sim$ N$(m,\frac{1}{\kappa\tau})$: $\mu|\tau,x \sim$ N$(\frac{\kappa m + n\bar{x}}{\kappa + n},\frac{1}{(\kappa+n)\tau})$.
- $\tau \sim$ Gamma(a,b): $\tau|\mu,x \sim$ Gamma$(a + \frac{n}{2},b + \frac{S}{2})$; with the $\mu|\tau$ prior: Gamma$(a + \frac{n+1}{2},b + \frac{S}{2} + \frac{\kappa}{2}(\mu - m)^2)$.
- $\sigma^2 \sim$ InvGamma(a,b): InvGamma$(a + \frac{n}{2},b + \frac{S}{2})$; prior $\mu|\sigma^2 \sim$ N$(m,\sigma^2/\kappa)$: InvGamma$(a + \frac{n+1}{2},b + \frac{S}{2} + \frac{\kappa}{2}(\mu - m)^2)$.
- **Two samples** x (n_1), y (n_2), common τ: Gamma$(a + \frac{n_1+n_2}{2},b + \frac{S_x+S_y}{2})$; μ_1 uses x only.
- **One μ, two sources:** $x_i \sim$ N$(\mu,\frac{1}{\tau})$ (n), $y_j \sim$ N$(c\mu,\frac{1}{d\tau})$ (m), $\mu \sim$ N$(\mu_0,\frac{1}{\kappa})$: $A = \kappa + (n + c^2dm)\tau$, $B = \kappa\mu_0 + \tau(\sum x_i + cd\sum y_j)$; $\tau|\cdot \sim$ Gamma$(a + \frac{n+m}{2},b + \frac{1}{2}[\sum(x_i - \mu)^2 + d\sum(y_j - c\mu)^2])$.
- **Hierarchical:** $y_{ij} \sim$ N$(\theta_j,\frac{1}{\tau})$, $\theta_j \sim$ N$(\mu,\frac{1}{\tau_\theta})$, $\mu \sim$ N$(m,\frac{1}{\kappa})$: $\theta_j|\cdot$ precision $n_j\tau + \tau_\theta$, mean $(n_j\tau\bar{y}_j + \tau_\theta\mu)$/prec (weight $n_j\tau$/prec on $\bar{y}_j$); $\mu|\cdot$ precision $\kappa + J\tau_\theta$, mean $(\kappa m + \tau_\theta\sum\theta_j)$/prec (no y); $\tau_\theta|\cdot \sim$ Gamma$(a + \frac{J}{2},b + \frac{1}{2}\sum(\theta_j - \mu)^2)$.
- **Bivariate prior**, $\Lambda = \Sigma^{-1}$, zero mean, $x_i \sim$ N(μ_1,σ^2): $-\frac{1}{2}(\Lambda_{11}\mu_1^2 + 2\Lambda_{12}\mu_1\mu_2 + \Lambda_{22}\mu_2^2)$ in the exponent gives $A = \Lambda_{11} + n/\sigma^2$, $B = n\bar{x}/\sigma^2 - \Lambda_{12}\mu_2$: $\mu_1|\mu_2,x \sim$ N$(B/A,1/A)$.
- **Change point:** $Y_i \sim$ Pois(λ_1), $i \leq m$; Pois(λ_2), $i > m$; Gamma priors; m uniform on $1..n$: $\lambda_1|\cdot \sim$ Gamma$(\alpha_1 + \sum_1^m y_i,\beta_1 + m)$; $\lambda_2|\cdot \sim$ Gamma$(\alpha_2 + \sum_{m+1}^n y_i,\beta_2 + n - m)$; $p(m|\cdot) \propto (\lambda_1/\lambda_2)^{\sum_1^m y_i}e^{(\lambda_2-\lambda_1)m}$.
- **Hidden count:** $X|p,N \sim$ bin(N,p), $p \sim$ Beta(a,b), $N \sim$ Pois(λ): $p|\cdot \sim$ Beta$(x+a,N-x+b)$; $N-x|\cdot \sim$ Pois$(\lambda(1-p))$.
- **Two groups, flat priors,** $p(\tau) \propto 1/\tau$: $\mu_1|\cdot \sim$ N$(\bar{x},\frac{1}{n\tau})$, $\mu_2|\cdot \sim$ N$(\bar{y},\frac{1}{m\tau})$, $\tau|\cdot \sim$ Gamma$(\frac{n+m}{2},\frac{1}{2}[\sum(x_i - \mu_1)^2 + \sum(y_j - \mu_2)^2])$.
- **Poisson two sources:** $y_i \sim$ Pois(λ) (n), $z_j \sim$ Pois($c\lambda$) (m): $\lambda|\cdot \sim$ Gamma$(a + \sum y_i + \sum z_j,b + n + cm)$.
- **Poisson hierarchy:** $\lambda_j|\beta \sim$ Gamma(α,β), $\beta \sim$ Gamma(c,d): $\lambda_j|\cdot \sim$ Gamma$(\alpha + \sum_i y_{ij},\beta + n_j)$; $\beta|\cdot \sim$ Gamma$(c + J\alpha,d + \sum\lambda_j)$.

MCMC diagnostics: (1) Converged? (2) How correlated?
Trace plots: run several chains from **overdispersed** starts (more variable than the target). Good: chains overlap, stable, "hairy caterpillar". Separate or drifting: not converged. Slow wander, "woolly snake": high autocorrelation. **Burn-in:** discard early iterations affected by the start; read length off the trace.
BGR: M chains, each length $L+T$; keep last T. **B** = sd of the pooled MT draws (between). **W** = average of the M within-chain sds (within). $\hat{R} = B/W < 1.05$: accept last T as stationary and pool. $\hat{R} \gg 1$: not converged. $T \to \infty$: $\hat{R} \to 1$. Exclude burn-in from $\hat{R}$.
Trace vs step size: tiny steps: smooth wander; huge steps: flat runs then jumps; good: fuzzy band. Accept near 1 and smooth: raise step. Low accept and plateaus: lower it. Gibbs accepts all, so use ACF, ESS, trace.
Not converged: run longer, better proposal. A pass is no proof: all starts may miss a region.
Autocorrelation: lag k: $\rho_k = \text{cor}(\theta^{(t)},\theta^{(t+k)})$; `acf(chain)`; `cor(chain[-n],chain[-1])` is $\hat{\rho}_1$. **Thinning:** keep every kth; simple, saves memory, throws away information. ESS never rises. `acf(x,plot=FALSE)$acf[2]` $= \hat{\rho}_1$ (entry 1 is lag 0); bars above the dashed band are real. `mean(diff(x)!=0)`: acceptance rate.
ESS: independent sample size with the same information: $n/(1 + 2\sum_i\rho_i) \approx n\frac{1-\rho_1}{1+\rho_1}$. Aim ESS > 4000. E.g. $n = 5000$, $\rho_1 = .9$: ESS ≈ 263. Naive SE too small by $\sqrt{(1+\rho_1)/(1-\rho_1)}$. n for a target ESS: target$\times\frac{1+\rho_1}{1-\rho_1}$.
Accuracy: Var$(\bar{h}) \approx \frac{\sigma^2}{n}(1 + 2\sum\rho_i)$: iid SE is too small. MC s.e.: iid $s/\sqrt{N}$; proportion $\sqrt{\hat{p}(1-\hat{p})/N}$; chain $s/\sqrt{\text{ESS}}$. Posterior sd does not shrink with N. **Blocking:** MC error = sd(block means)$/\sqrt{Q}$; want $\leq 5\%$ of posterior sd. Same as ESS ≥ 400.

Maid Mast Codes PT4: MCMC

- `prob<-dtarget(prop)/dtarget(cur); if(runif(1)<prob) accept`: symmetric MH. `is.nan` guard. `proposal<-rnorm(2,chain[i,],sd)`: random walk.
- `acceptance<-1-mean(duplicated(chain[-(1:burnIn),]))`: after burn-in. `chain[-(1:burnIn),]` drops burn-in.
- `la1<-rgamma(1,a1+cfreq[M],b1+M)`: Gamma(shape, rate); `cfreq<-cumsum(freq)` gives $\sum_1^m y_i$.
- `M<-sample(1:n,1,prob=(la1/la2)^cfreq*exp((la2-la1)*(1:n)))`: discrete conditional.
- `rnorm(1,mu1+(x2-mu2)*s12/s2,sqrt(s1-s12^2/s2))`: Gibbs, sd not var.
- `cumsum(x)/1:n`: running mean, should flatten. `mean(th>0)`: posterior probability. `s[seq(1,n,by=k),]`: thin.
- `1/rgamma(1,a,b)`: InvGamma draw. `rnorm(1,m,sqrt(1/A))`: sd is the root of the variance.

VARIATIONAL INFERENCE

Variational inference: turns inference into optimisation: find $q^* \in D$ nearest the posterior. Faster for large data and complex models, at a cost in accuracy.
KL: $KL(q\|p(\theta|y)) = \int q(\theta)[\log q(\theta) - \log p(\theta|y)]d\theta \geq 0$. Not symmetric. Penalises q for mass where p is tiny, not for missing mass, so q sits inside p.
ELBO: $p(\theta|y)$ unknown, but $KL = -ELBO + \log p(y)$ with

$$ELBO(q) = -\int q[\log q - \log p(\theta)p(y|\theta)]d\theta$$

$= \mathbb{E}_q[\log p(\theta,y)] - \mathbb{E}_q[\log q]$. $\log p(y)$ is free of q, so **min KL = max ELBO**; $KL \geq 0$ gives $\log p(y) \geq ELBO$. $\log p(y) = ELBO + KL$: the gap is the KL.
Mean field: $q(\theta) = \prod_j q_j(\theta_j)$: parameters independent under q. An assumption on the approximation, not on the model.
CAVI: update one factor at a time, others fixed: $q_j^*(\theta_j) \propto \exp\{\mathbb{E}_{-j}[\log p(\theta,y)]\}$. ELBO rises to a local optimum (a drop means a bug). Stop when ELBO change is small. Sensitive to initial values: run several, keep highest ELBO. It updates distributions; it draws no samples. Checks: ELBO never falls, stays $\leq \log p(y)$.
Recipe: (1) $\log p(\theta,y)$ = log likelihoods + log priors. (2) Keep terms with θ_j. (3) Take $\mathbb{E}$ over the others (linear). (4) Collect powers, name the kernel. Same kernel as Gibbs, with terms in other parameters replaced by their expectations.
Use: $\mathbb{E}(x-\mu)^2 = (x - \mathbb{E}\mu)^2 + V\mu = \sigma^{2*} + \mu^{*2} - 2x\mu^* + x^2$; $\mathbb{E}\mu^2 = (\mathbb{E}\mu)^2 + V\mu$; $\mathbb{E}\tau = a^*/b^*$; $\mathbb{E}\log\tau = \psi(a^*) - \log b^*$; $\mathbb{E}[1/\sigma^2] = a^*/b^*$. q_μ uses $\mathbb{E}[1/\sigma^2]$, not $1/\mathbb{E}\sigma^2$.
Normal-Gamma: $x_i \sim$ N$(\mu,\frac{1}{\tau})$, $\mu|\tau \sim$ N$(\mu_0,\frac{1}{\lambda_0\tau})$, $\tau \sim$ Gamma(a_0,b_0):
$q_\mu^* =$ N(μ^*,σ^{2*}): $\mu^* = \frac{\lambda_0\mu_0 + n\bar{x}}{\lambda_0 + n}$, $\sigma^{2*} = \frac{1}{(\lambda_0+n)\mathbb{E}\tau}$.
$q_\tau^* =$ Gamma(a^*,b^*): $a^* = a_0 + \frac{n+1}{2}$, $b^* = b_0 + \frac{1}{2}\sum_i\mathbb{E}(x_i - \mu)^2 + \frac{\lambda_0}{2}\mathbb{E}(\mu - \mu_0)^2$.
Only σ^{2*} and b^* change between iterations. **Variance form:** $q_{\sigma^2}^*$ is InvGamma with the same two parameters.
Independent priors $\mu \sim$ N$(m,\frac{1}{\kappa})$, $\tau \sim$ Gamma(a,b): $q_\mu^* =$ N$(\frac{\kappa m + n\bar{x}\mathbb{E}\tau}{\kappa + n\mathbb{E}\tau},\frac{1}{\kappa + n\mathbb{E}\tau})$; $q_\tau^* =$ Gamma$(a + \frac{n}{2},\ b + \frac{1}{2}[\sum(x_i - \mathbb{E}\mu)^2 + nV\mu])$. Shape $a + \frac{n}{2}$, not $\frac{n+1}{2}$; here both factors move each round. **Two sources,** y_j variance $d\sigma^2$: $\mathbb{E}Q = \sum(x_i - \mathbb{E}\mu)^2 + nV\mu + \frac{1}{d}[\sum(y_j - \mathbb{E}\mu)^2 + mV\mu]$, shape $a + \frac{n+m}{2}$.
Exact: $\mu|\tau,x \sim$ N$(\frac{\lambda_0\mu_0 + n\bar{x}}{\lambda_0+n},\frac{1}{(\lambda_0+n)\tau})$, $\tau|x \sim$ Gamma$(a_0 + \frac{n}{2},b_0 + \frac{1}{2}\sum(x_i - \bar{x})^2 + \frac{n\lambda_0(\bar{x}-\mu_0)^2}{2(\lambda_0+n)})$. Var$(\mu|x) = \frac{b_n}{(n+\lambda_0)(a_n-1)}$.
Bivariate Normal target: $\Lambda = \Sigma^{-1}$: $q_1^* =$ N$(\mu_1 - \frac{\Lambda_{12}}{\Lambda_{11}}(\mathbb{E}x_2 - \mu_2),\frac{1}{\Lambda_{11}})$. $1/\Lambda_{11} = \sigma_1^2 - \sigma_{12}^2/\sigma_2^2 < \sigma_1^2$: the conditional variance. Means right, but VI underestimates variance and ignores correlation; worse as correlation rises. Error in the means shrinks by ρ^2 a round: strong correlation is slow too.

MCMC vs VI: MCMC simulates the density, is heavier, asymptotically exact. VI approximates it, is faster, suits large data and model exploration, no guarantee, underestimates variance.

EXAM CHECKS

- Define P_{ik}; state which θ^0.
- State H_0, statistic, df, decision, sentence in context.
- CI on η first, transform last.
- $\hat{p}$, $\hat{\mu}$ on response scale.
- p includes intercept.
- $\hat{\phi}$: big model for F, own model for SE.
- Rate vs scale; variance vs sd in N.
- Name distribution and every parameter.
- 4 dp inside, round once.

PAGE 3: CUT FROM PAGES 1 AND 2

Holding page. Exam notes are one A4 sheet, double-sided, so this does not go in as it stands.

PREREQUISITES

Confidence intervals: **Pivot:** function of data and θ with fully known distribution.

Want	95% CI
μ, σ known	$\bar{x} \pm z\,\sigma/\sqrt{n}$
μ, σ unk.	$\bar{x} \pm t_{n-1}\,s/\sqrt{n}$
$\mu_X - \mu_Y$, equal var	$\bar{x} - \bar{y} \pm t_{n+m-2}\,s_p\sqrt{1/n + 1/m}$
p	$\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$
σ^2 (χ^2_{n-1} c)	$[(n-1)s^2/c_{.975},\ (n-1)s^2/c_{.025}]$
New X^* (PI)	$\bar{x} \pm t_{n-1}\,s\sqrt{1+1/n}$

Tests, estimators: H_1: effect. **Test statistic:** known distribution under H_0. Statistical ≠ practical significance. **Unbiased:** Bias 0. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ for Normal data. **Model terms:** Test a factor by one LRT on $L-1$ df, not its separate z values. Observational data show **association, not causation**. A predictor that flips sign is “not significant in the presence of the others”.

Bivariate Normal pdf: $f(x) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x-\mu)\}$.

More cutoffs: $\chi^2_{df}(.95)$: 8: 15.51, 25: 37.65. $\chi^2_{df}(.99)$: 2: 9.21, 3: 11.34. $\Phi(1) = .84$, $\Phi(2) = .98$. $t_{60}(.975) = 2.00$.

LIKELIHOOD & BINOMIAL REGRESSION

More MLE rows:

iid $X_i \sim$	MLE	$I(\theta)$
Bernoulli(p)	$\bar{x}$	$n/(p(1-p))$
$\theta x^{\theta-1}$ on (0,1)	$-n/\sum \log x_i$	n/θ^2
$\frac{x}{\theta} e^{-x^2/2\theta}$	$\sum x_i^2/2n$	n/θ^2

Logit information, one predictor:

$$I(\beta) = \begin{pmatrix} \sum m_i p_i q_i & \sum m_i x_i p_i q_i \\ \sum m_i x_i p_i q_i & \sum m_i x_i^2 p_i q_i \end{pmatrix},\ q_i = 1 - p_i$$

Probit: $I = X^\top W X$, $W_{ii} = m_i \varphi(\eta_i)^2/[\Phi(\eta_i)(1 - \Phi(\eta_i))]$.

GLMs: DEVIANCE, IWLS, DIAGNOSTICS

Why the deviance (Poisson): $l_S = \sum[y_i \log y_i - y_i - \log y_i!]$, $\hat{l} = \sum[y_i \log \hat{\mu}_i - \hat{\mu}_i - \log y_i!]$; double the gap.

IWLS, other families in full:

Family, link	z_i	W_{ii}
Poisson, $\sqrt{\ }$	$\eta/2 + y/(2\eta)$	4
Binom, cloglog	$\eta + \frac{y-\mu}{m(1-p)L}$	$m(1-p)L^2/p$
Gamma, $1/\mu$	$\eta - (y-\mu)/\mu^2$	μ^2
Gamma, identity	y	$1/\mu^2$
IG, $1/\mu$	$\eta - (y-\mu)/\mu^2$	μ

Cloglog: $L = e^\eta = -\log(1-p)$.

OVERDISPERSION & QUASI-LIKELIHOOD

Quasi-score in full: $u_i = \frac{Y_i - \mu_i}{\phi v(\mu_i)}$: $\mathbb{E}u = 0$, Var$u = \frac{1}{\phi v}$, $-\mathbb{E}\frac{\partial u}{\partial \mu} = \frac{1}{\phi v} = \mathbb{E}u^2$, same as $\partial \log L/\partial \mu$.

READING R OUTPUT

- **Deviance Residuals** (old R): want median near 0, roughly symmetric.
- **Std. Error:** $\sqrt{V_{jj}} = \sqrt{[I(\hat{\beta})^{-1}]_{jj}}$.
- **Interaction tested twice:** z from Fisher information ~ $\mathrm{N}(0,1)$, and $\Delta D \sim \chi^2_1$. Prefer ΔD.

MIXTURES & EM

- Exp(λ), Exp(2λ): $P_{i1} = \pi/[\pi + 2(1-\pi)e^{-\lambda x_i}]$.
- $\mathrm{N}(\mu,1)$, $\mathrm{N}(2\mu,4)$: $\hat{\mu} = \frac{1}{n}\sum(P_{i1} + \frac{1}{2}P_{i2})x_i$.
- $K = 3$: $N_k/\pi_k = c$ for all k, sum gives $c = n$.
- **Is it a maximum?** $Q''_\mu = -N_k/\sigma_k^2$; $Q''_\pi = -N_1/\pi^2 - N_2/(1-\pi)^2$; $Q''_{\sigma^2} = -N_k/2\hat{\sigma}_k^4$.
- **Overlap:** P_{ik} less decisive, EM slower and more start-sensitive. Do not compare log-lik across different data.
- `length(log_liks)` = iterations +1.

BAYES

- **In λ:** Poisson likelihood is $\lambda^{\sum y_i} e^{-n\lambda}$, Exp is $\lambda^n e^{-\lambda \sum y_i}$: Gamma kernels. Exp(rate b) = Gamma$(1,b)$.
- **Exp-Gamma mean:** weight $\frac{b}{b+\sum x_i}$ on a/b, rest on $n/\sum x_i$.
- **Predictive t:** known mean μ_0, $\tau|x \sim$ Gamma(a^*,b^*): 3-parameter t, $\nu = 2a^*$, location μ_0, scale² $s^2 = b^*/a^*$, pdf $\propto (1 + \frac{(x-\mu_0)^2}{\nu s^2})^{-(\nu+1)/2}$.
- **Spike prior:** $p(x|\gamma) = \int p(x|\mu)p(\mu|\gamma)d\mu$.
- **Conjugate exp family:** $f = \alpha(x)e^{x\theta + \beta(\theta)}$, prior $\propto e^{\gamma\theta + \delta\beta(\theta)}$: posterior $(\gamma + x, \delta + 1)$.
- **Joint prior $\mu|\tau$, τ:** exact posterior (VI box), so sample τ then μ, no MCMC.

SIMULATION

Why rejection works: (X,Y) is uniform under kh (density $1/k$); points kept are uniform under f, so $X \sim f$.

MCMC

Why it works (not examinable): detailed balance $\pi(\theta)p(\theta \to \theta') = \pi(\theta')p(\theta' \to \theta)$ makes π stationary; aperiodic, irreducible: limiting; ergodic: averages converge.

MH derivation, $x \sim \mathrm{N}(\mu,1)$, $\mu \sim \mathrm{N}(0,1)$, random walk: $\log\frac{\pi(\mu')}{\pi(\mu)} = -\frac{1}{2}[(x-\mu')^2 + \mu'^2] + \frac{1}{2}[(x-\mu)^2 + \mu^2]$; $A = \min(1, e^{\text{this}})$.

Gibbs μ derivation: $p(\mu|\tau,x) \propto \exp\{-\frac{\tau}{2}\sum(x_i - \mu)^2 - \frac{\kappa}{2}(\mu - m)^2\} \propto \exp\{-\frac{1}{2}(n\tau + \kappa)\mu^2 + (\tau n\bar{x} + \kappa m)\mu\}$: $A = n\tau + \kappa$, $B = \tau n\bar{x} + \kappa m$.

Exp change point: $\lambda_1|\cdot \sim$ Gamma$(a_1 + M, b_1 + S_M)$, $\lambda_2|\cdot \sim$ Gamma$(a_2 + n - M, b_2 + S_n - S_M)$, $S_m = \sum_1^m y_i$; $p(m|\cdot) \propto (\lambda_1/\lambda_2)^m e^{-(\lambda_1 - \lambda_2)S_m}$.

Discrete $\mu \in \{0,1\}$: $P(\mu = 1|\theta) = f(\theta,1)/[f(\theta,0) + f(\theta,1)]$.

VARIATIONAL INFERENCE

CAVI τ derivation: $\log q^*_\tau = \mathbb{E}_\mu[\log p(x|\mu,\tau) + \log p(\mu|\tau) + \log p(\tau)] + c = (\frac{n}{2} + \frac{1}{2} + a_0 - 1)\log\tau - \tau[b_0 + \frac{1}{2}\sum\mathbb{E}(x_i - \mu)^2 + \frac{\lambda_0}{2}\mathbb{E}(\mu - \mu_0)^2] + c$: power of $\log\tau$ is $a^* - 1$, coefficient of $-\tau$ is b^*.

ELBO, Normal-Gamma: $\mathbb{E}[\log p(x,\mu,\tau)] - \mathbb{E}\log q_\mu - \mathbb{E}\log q_\tau$, with $\mathbb{E}\log q_\mu = -\frac{1}{2}\log\sigma^{2*} + c$, $\mathbb{E}\log q_\tau = \log b^* - \log\Gamma(a^*) + (a^* - 1)\psi(a^*) - a^*$.

ELBO, bivariate target, up to a constant, $q_i = \mathrm{N}(m_i, s_i^2)$: $-\frac{1}{2}[\Lambda_{11}(s_1^2 + (m_1 - \mu_1)^2) + \Lambda_{22}(s_2^2 + (m_2 - \mu_2)^2) + 2\Lambda_{12}(m_1 - \mu_1)(m_2 - \mu_2)] + \frac{1}{2}\log s_1^2 + \frac{1}{2}\log s_2^2$.

Bivariate prior (Gibbs row): $q_1^* = \mathrm{N}(B/A, 1/A)$ with μ_2 replaced by $\mathbb{E}\mu_2$.

Variance form: $\sigma^2 \sim$ InvGamma(a_0, b_0), $\mu|\sigma^2 \sim \mathrm{N}(\mu_0, \sigma^2/\lambda_0)$.

EXAM CHECKS

- N(mean, variance), Gamma(shape, rate), InvGamma(shape, scale).